



计算机网络实验报告

警告

1. 实验报告如有雷同，雷同各方当次实验成绩均以 0 分计。
2. 当次小组成员成绩只计学号、姓名登录在下表中的。
3. 在规定时间内未上交实验报告的，不得以其他方式补交，当次成绩按 0 分计。
4. 实验报告文件以 PDF 格式提交。

院系	计算机学院	班 级	计科 1 班	组长	郝裕玮
学号	18329015	18325071	19335153		
学生	郝裕玮	张闯	马淙升		
实验分工					
郝裕玮	张闯	马淙升			
共同完成实验与实验报告	共同完成实验与实验报告	共同完成实验与实验报告			

【实验题目】访问控制列表（ACL）实验。

【实验目的】

1. 掌握标准访问列表规则及配置。
2. 掌握扩展访问列表规则及配置。
3. 了解标准访问列表和扩展访问列表的区别。

【实验内容】

完成教材实例 8-4（P296），请写出步骤 1 安装与建立 FTP、WEB 的步骤，并完成 P297~P298 的测试要求。

【实验要求】

重要信息需给出截图，注意实验步骤的前后对比。

【实验记录】(如有实验拓扑请自行画出)

【实验拓扑】

某公司的网络中使用 1 台路由器提供子网间的互连。子网 192.168.1.0/24 为公司员工主机所在的网段,其中公司经理的主机地址为 192.168.1.254/24;子网 10.1.1.0/24 为公司服务器网段,其中有 2 台服务器、1 台 WWW 服务器(10.1.1.100/24)和 1 台 FTP 服务器(10.1.1.200/24)。现在要实现基于时间段的访问控制,使公司员工只有在正常上班时间(周一至周五 9:00~18:00)可以访问 FTP 服务器,并且只有在下班时间才能访问 WWW 服务器,而经理的主机可以在任何时间访问这 2 台服务器。

本实验的拓扑结构如图 8-8 所示。

实验拓扑图如下所示：

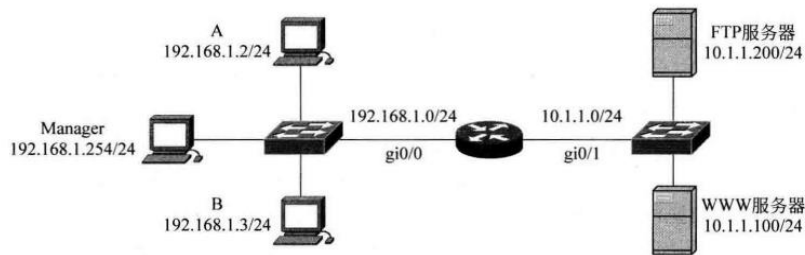


图 8-8 基于时间 ACL 的实验拓扑

【实验设备】

路由器 1 台,计算机 5 台(其中 2 台作为 WWW 服务器和 FTP 服务器)。

【实验步骤】

分析：与此前的许多实验做法不同，实验结果不能简单地通过 ping 验证，因为本实验并不禁止 ICMP 包。按照实验要求，必须搭建 FTP 和 WWW 服务器。

对于服务器的搭建我们可以选择安装第三方专业软件，例如 FTP 服务器可选择安装 Filezilla Server, WWW 服务器可选择 Apache Server。FTP 服务器需至少创建一个用户名和口令。

同时，本实验的预期目标是通过配置 ACL 来实现基于时间段的访问控制，使公司员工只有在正常上班时间(周一至周五 9:00~18:00)可以访问 FTP 服务器,并且只有在下班时间才能访问 WWW 服务器，而经理的主机可以在任何时间访问这 2 台服务器。

步骤 1:

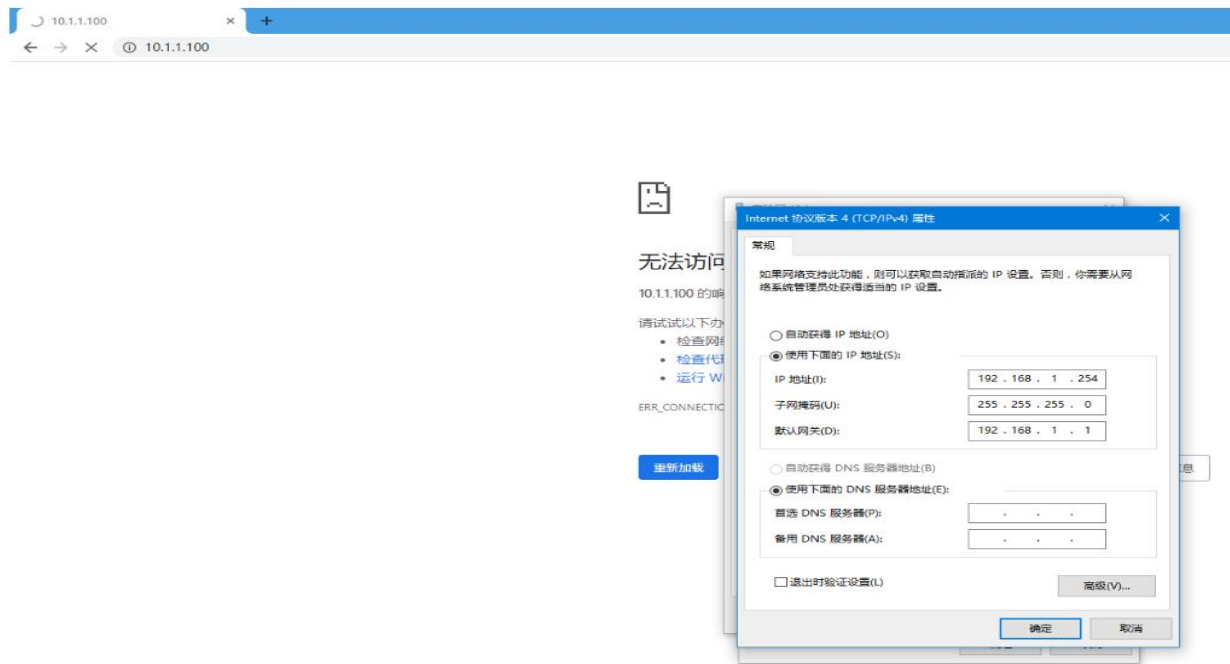
(1)配置 3 台计算机(A、B 和 Manager)的 IP 地址、子网掩码、网关。

根据拓扑图可得以下表格：

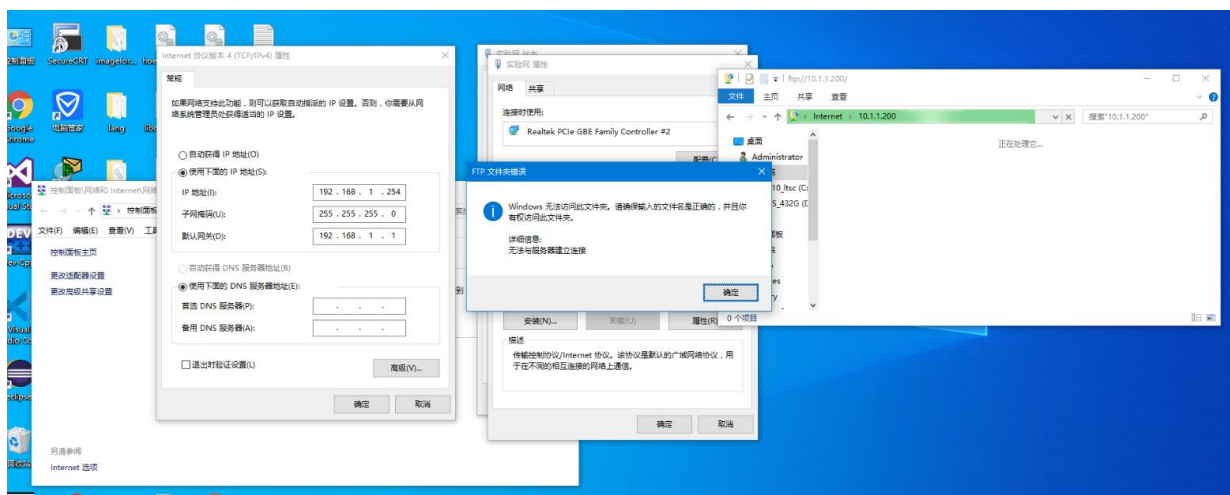
	A	B	Manager
IP 地址	192.168.1.2	192.168.1.3	192.168.1.254
子网掩码	255.255.255.0	255.255.255.0	255.255.255.0
网关	192.168.1.1	192.168.1.1	192.168.1.1

(2)检查计算机与服务器的连通性。

计算机通过 Web 访问服务器：



计算机通过 FTP 访问服务器：



由上可知，此时计算机与服务器无法连通。

(3)在服务器上安装 FTP 服务器和 WWW 服务器。FTP 服务器需至少创建一个用户名和口令。

安装配置 FTP 服务器的过程为：

首先按照安装指南建立默认的服务器 localhost（端口 14147，密码 123456）

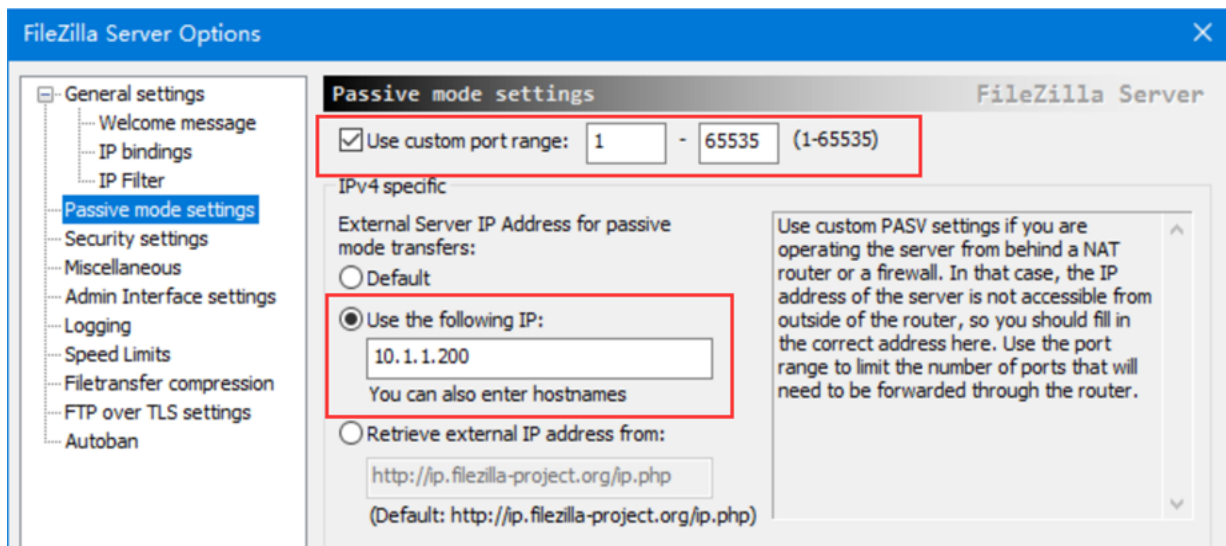
进入后登录成功但报错：

```
FileZilla Server 0.9.60 beta
Copyright 2001-2016 by Tim Kosse (tim.kosse@filezilla-project.org)
https://filezilla-project.org/
Connecting to server localhost:14147...
Connected, waiting for authentication
Logged on
You appear to be behind a NAT router. Please configure the passive mode settings and forward a range of ports in your router.
Warning: FTP over TLS is not enabled, users cannot securely log in.
```

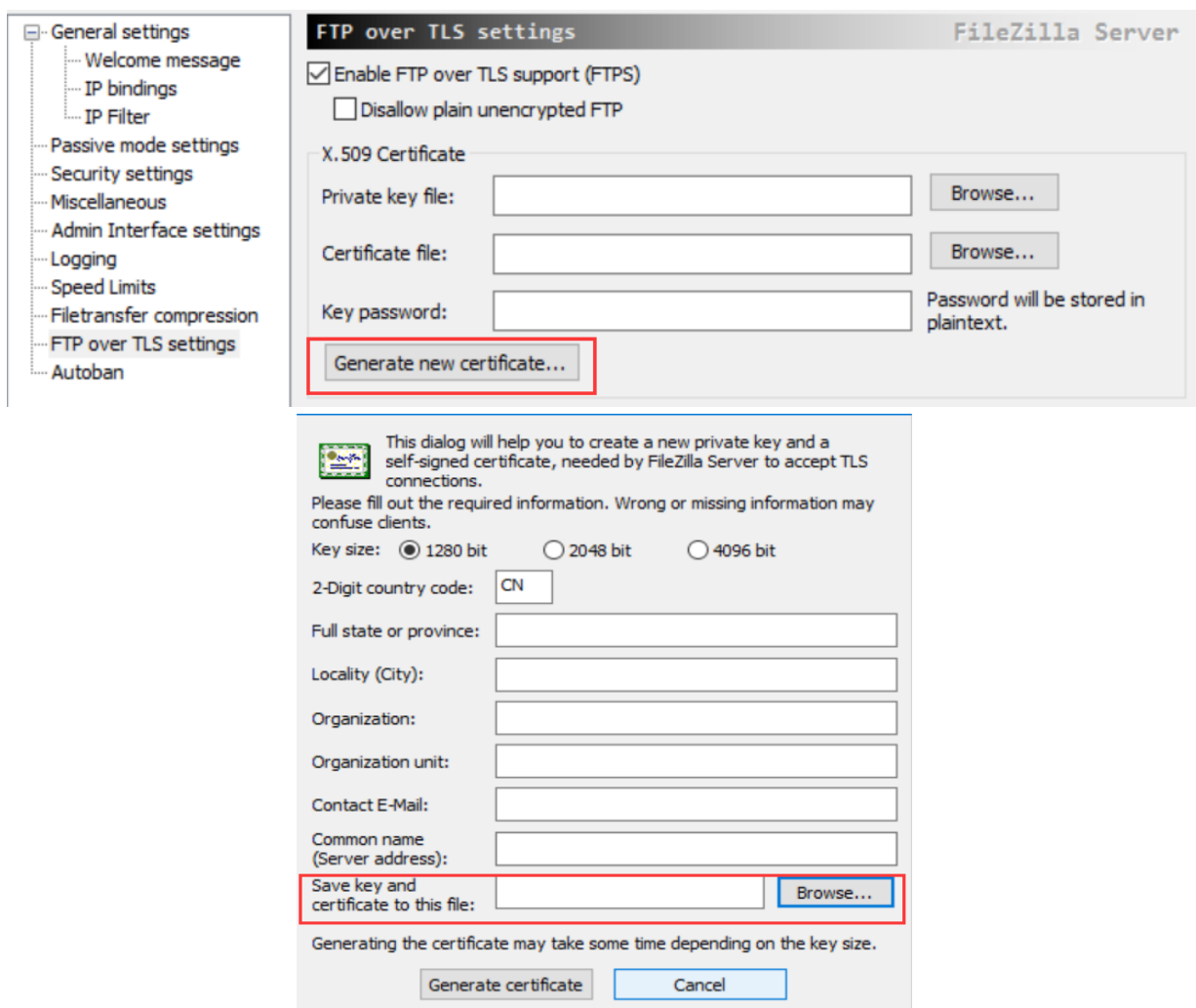


查阅资料后得知，需修改以下两部分内容：

① 修改接收端口范围和 IP 地址：



② 生成新的证书：



除红色方框以外的内容均可随意填写，红色方框内的文件位置最好定位在 Filezilla 的安装目录下。



修改过后不再报错:

```
FileZilla Server 0.9.60 beta
Copyright 2001-2016 by Tim Kosse (tim.kosse@filezilla-project.org)
https://filezilla-project.org/
Connecting to server localhost:14147...
Connected, waiting for authentication
Logged on
You appear to be behind a NAT router. Please configure the passive mode settings and forward a range of ports in your router.
Warning: FTP over TLS is not enabled, users cannot securely log in.
Retrieving settings, please wait...
Done retrieving settings
Sending settings, please wait...
Done sending settings.
Warning: FTP over TLS is not enabled, users cannot securely log in.
Retrieving settings, please wait...
Done retrieving settings
```

同时再增加登录用户，设置用户名，密码和共享文件夹属性:

The image shows two screenshots of the FileZilla Server Users configuration window. The top screenshot shows the 'Account settings' tab for a user named 'localhost'. The 'Enable account' and 'Password' fields are checked and filled. The 'Group membership' is set to '<none>'. The 'Bypass userlimit of server' checkbox is unchecked, and the 'Maximum connection count' and 'Connection limit per IP' are both set to 0. The 'Force TLS for user login' checkbox is also unchecked. The bottom screenshot shows the 'Shared folders' tab for the same user. The 'Directories' list contains 'H C:\Users\Administ...'. The 'Aliases' list is empty. The 'Files' section has 'Read' checked, and 'Write', 'Delete', and 'Append' are unchecked. The 'Directories' section has 'Create', 'Delete', and 'List' unchecked, and '+ Subdirs' checked. The 'Set as home dir' button is visible. The 'Users' list on the right shows 'localhost'.

至此，FTP 服务器配置完成



安装配置 WWW 服务器的过程为:

配置 Apache server 到官网下载 Apache HTTP server, 解压至电脑根目录下。打开 conf 文件夹, 打开 httpd.conf 文件, 打开 HTTP 服务 (查找 ServerName, 删去前面的注释), 查看端口, 即 listen 行, 默认端口为 80, 如有需要可以修改。

打开终端, 进入 Apache 下的 bin 目录, 执行 httpd.exe -k install -n “Apache24” 指令, 安装 HTTP 服务。

之后运行 bin 下的 ApacheMonitor.exe 可执行文件, 即可运行 Web 服务器。

步骤 2:路由器的基本配置。

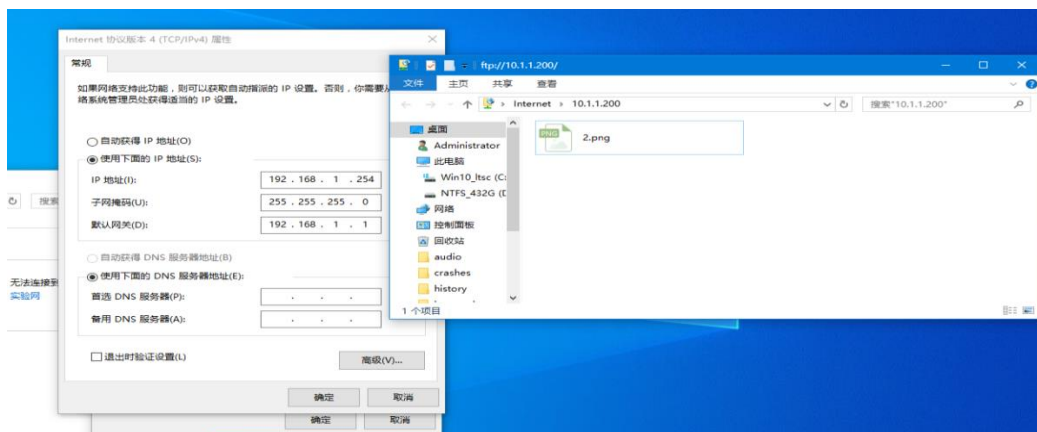
```
Router#configure terminal
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#exit
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 10.1.1.1 255.255.255.0
Router(config-if)#exit
```

```
Ruijie>enable
Ruijie#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Ruijie(config)#show ip router
% Invalid input detected at '^' marker.
Ruijie(config)#show ip route
Codes: C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default
Gateway of last resort is no set
C 192.168.1.0/24 is directly connected, GigabitEthernet 0/0
C 192.168.1.1/32 is local host.
Ruijie(config)#interface gi 0/0
Ruijie(config-if-GigabitEthernet 0/0)#ip address 192.168.1.1 255.255.255.0
Ruijie(config-if-GigabitEthernet 0/0)#exit
Ruijie(config)#interface gi 0/1
Ruijie(config-if-GigabitEthernet 0/1)#ip address 10.1.1.1 255.255.255.0
Ruijie(config-if-GigabitEthernet 0/1)#exit
```

步骤 3:验证当前配置。

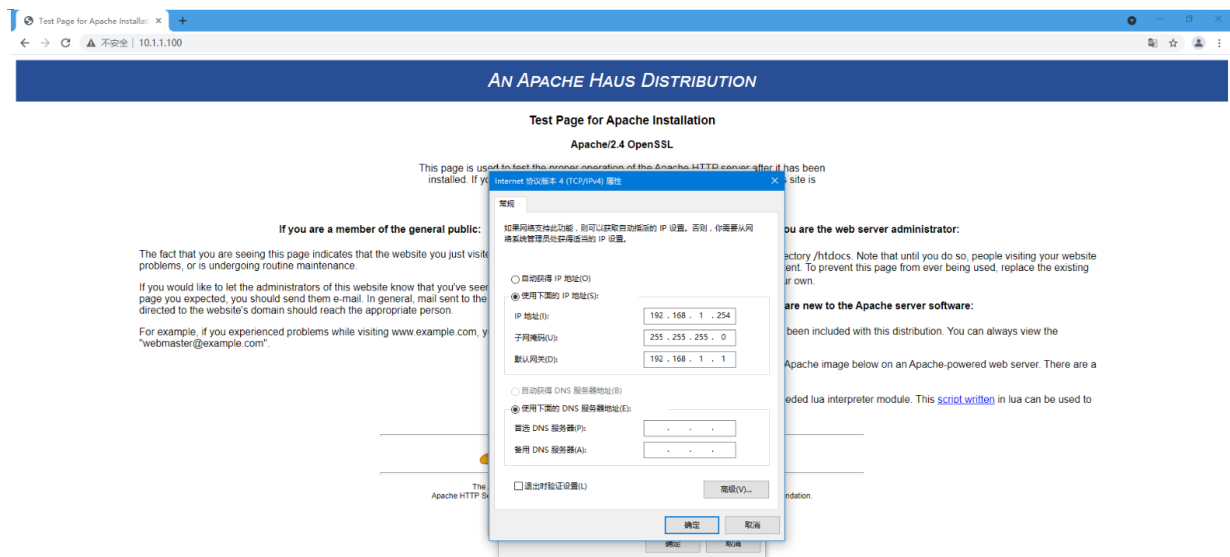
(1)验证主机与服务器的连通性。

对于 FTP 服务器:





对于 WWW 服务器:

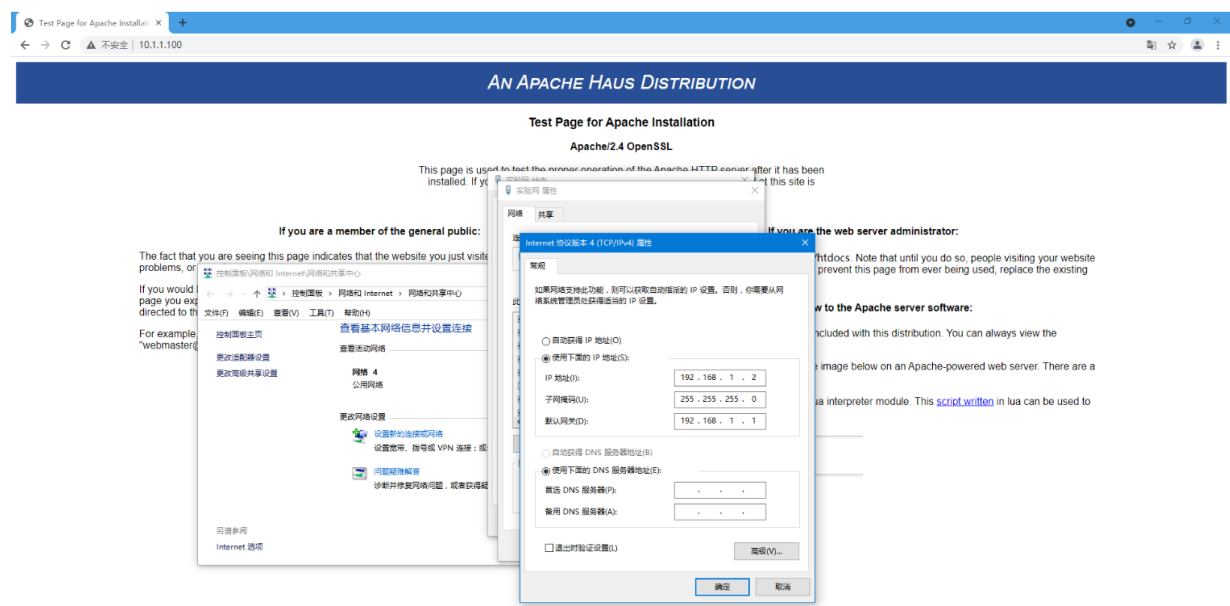


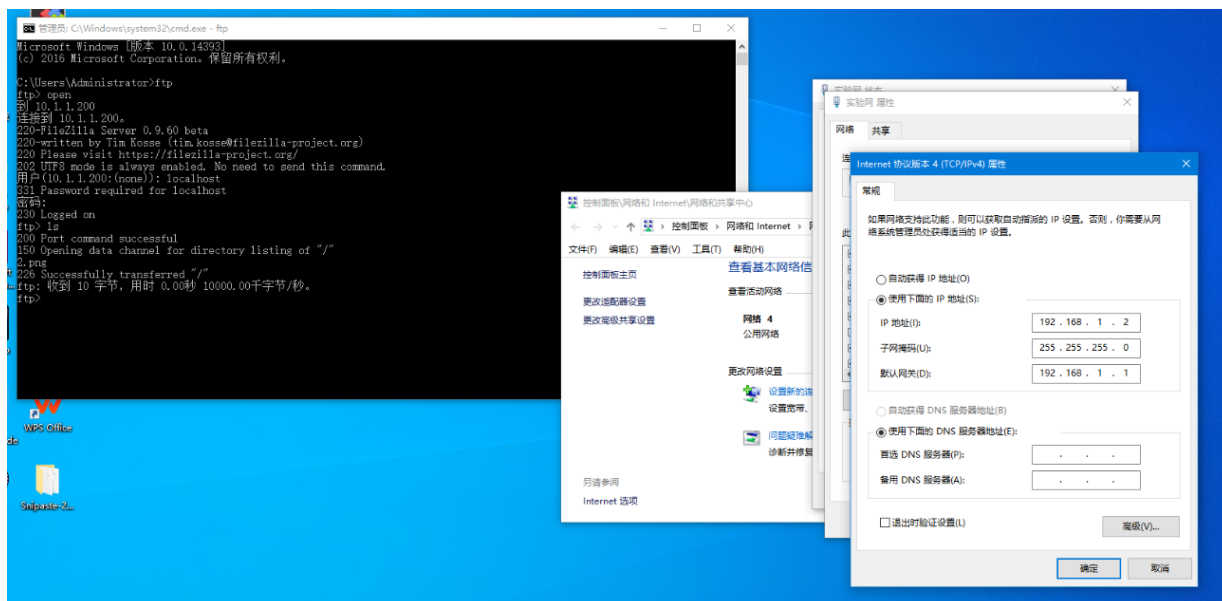
综合上述两图可知, 此时主机可以访问 FTP 服务器和 WWW 服务器。

(2) 经理机和员工机能否登录 FTP 服务器? 通过 `http://10.1.1.100` 能否访问 WWW 服务器? 判断目前结果是否达到预期目标, 并说明原因。

经理机访问 FTP 服务器和 WWW 服务器的结果已放在上面的(1)中 (均能访问)。

员工机访问 FTP 服务器和 WWW 服务器的结果如下所示 (A、B 对称, 所以这里仅展示 A 即可):





综上可知，经理机和员工机此时均能访问 FTP 服务器和 WWW 服务器。

步骤 4:配置时间段。

定义正常上班的时间段。

```
Router(config)#time-range work-time
Router(config-time-range)#periodic weekdays 09:00 to 18:00
Router(config-time-range)#exit
```

```
Ruijie(config)#time-range work-time
Ruijie(config-time-range)#periodic weekdays 09:00 to 18:00
Ruijie(config-time-range)#exit
```

步骤 5:配置 ACL。

配置 ACL 并应用时间段,以实现需求中基于时间段的访问控制。

```
Router(config)#ip access-list extended accessctrl
Router(config-ext-nacl)#permit ip host 192.168.1.254 10.1.1.0 0.0.0.255
!允许经理的主机在任何时间访问 2 台服务器
Router(config-ext-nacl)#permit tcp 192.168.1.0 0.0.0.255 host 10.1.1.200 eq ftp
time-range work-time
Router(config-ext-nacl)#permit tcp 192.168.1.0 0.0.0.255 host 10.1.1.200 eq ftp-
data time-range work-time
!只允许员工的主机在上班时间访问 FTP 服务器
Router(config-ext-nacl)#deny tcp 192.168.1.0 0.0.0.255 host 10.1.1.100 eq www
time-range work-time
!不允许员工的主机在上班时间访问 www 服务器
Router(config-ext-nacl)#permit tcp 192.168.1.0 0.0.0.255 host 10.1.1.100 eq www
!允许员工访问 www 服务器,但是仅当系统时间不在定义的时间段范围内时,才会执行此规则
Router(config-ext-nacl)#exit
```

```
Ruijie(config)#ip access-list extended accessctrl
Ruijie(config-ext-nacl)#permit ip host 192.168.1.254 10.1.1.0 0.0.0.255
Ruijie(config-ext-nacl)#permit tcp 192.168.1.0 0.0.0.255 host 10.1.1.200 eq ftp time-range work-time
Ruijie(config-ext-nacl)#permit tcp 192.168.1.0 0.0.0.255 host 10.1.1.200 eq ftp-data time-range work-time
Ruijie(config-ext-nacl)#deny tcp 192.168.1.0 0.0.0.255 host 10.1.1.100 eq www time-range work-time
Ruijie(config-ext-nacl)#permit tcp 192.168.1.0 0.0.0.255 host 10.1.1.100 eq www
Ruijie(config-ext-nacl)#exit
```




步骤 6:应用 ACL。

将 ACL 应用到端口 0/0 的输入方向。

```
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip access-group accessctrl in
Router(config-if)#end
```

```
Ruijie(config)#interface gi 0/0
Ruijie(config-if-GigabitEthernet 0/0)#ip access-group accessctrl in
Ruijie(config-if-GigabitEthernet 0/0)#end
```

步骤 7:验证测试。

在使用基于时间的 ACL 时,要保证设备(路由器或交换机)的系统时间的准确性,因为设备是根据自己的系统时间(而不是主机时间)判断当前时间是否在时间段范围内。可以在特权模式下使用 `show clock` 命令查看当前系统时间,并使用 `clock set` 命令调整系统时间。通过调整设备的系统时间实现在不同时间段测试 ACL 是否生效。

本实验分别做下列测试:

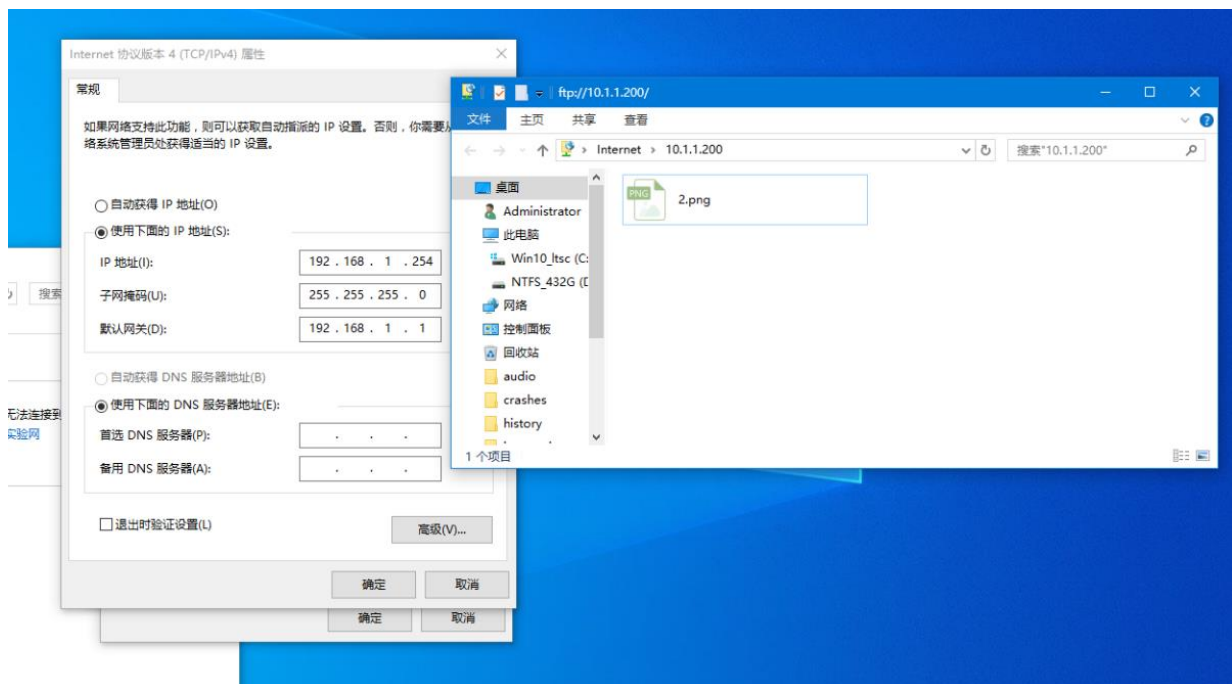
(1)查看路由器的系统时间:使用 `show clock` 命令判断当前时间段。

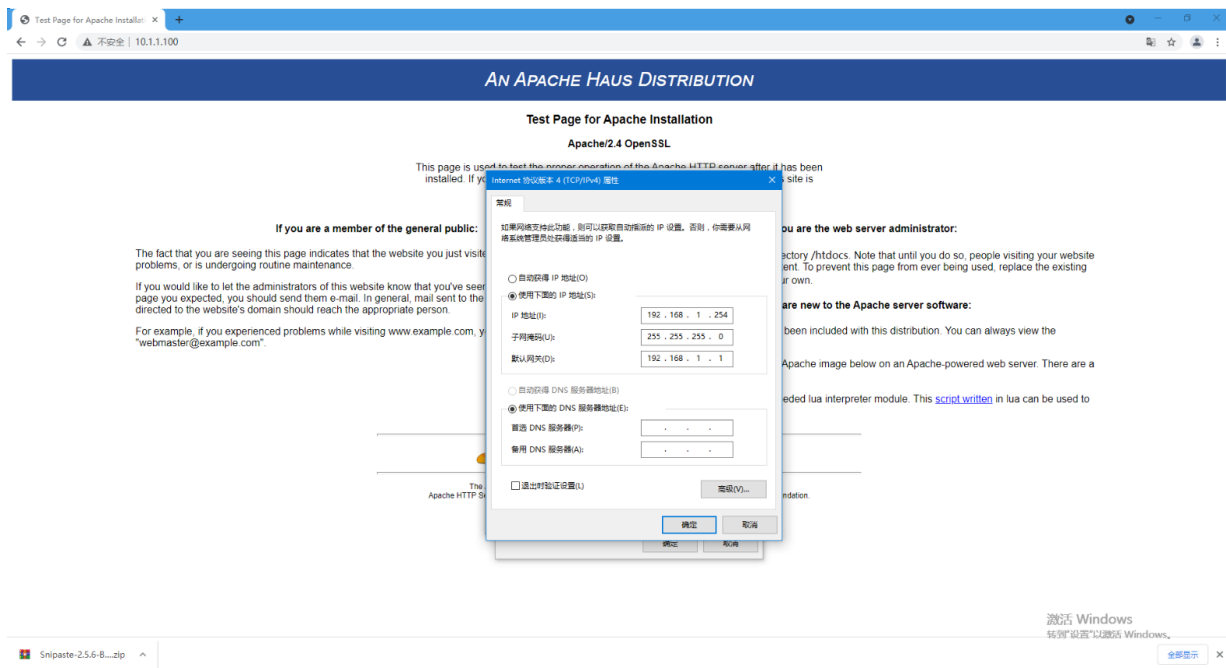
```
Ruijie#show clock
17:15:37 UTC Sat, Jun 5, 2021
Ruijie#clock set 12:00:00 6 4 2021
Ruijie#Jun 4 12:00:00: %SYS-6-CLOCKUPDATE: System clock has been updated to 12:00:00 UTC Fri Jun 4 2021.
```

在经过 `clock set` 之后易知当前时间为工作时间。

(2)经理的主机 Manager 使用步骤 1 建立的用户名登录 FTP 服务器,并通过 `http://10.1.1.100` 访问 WWW 服务器,在设定时间段内是否能登录和访问?

对于经理机,FTP 服务器和 WWW 服务器在设定时间段内均可访问:

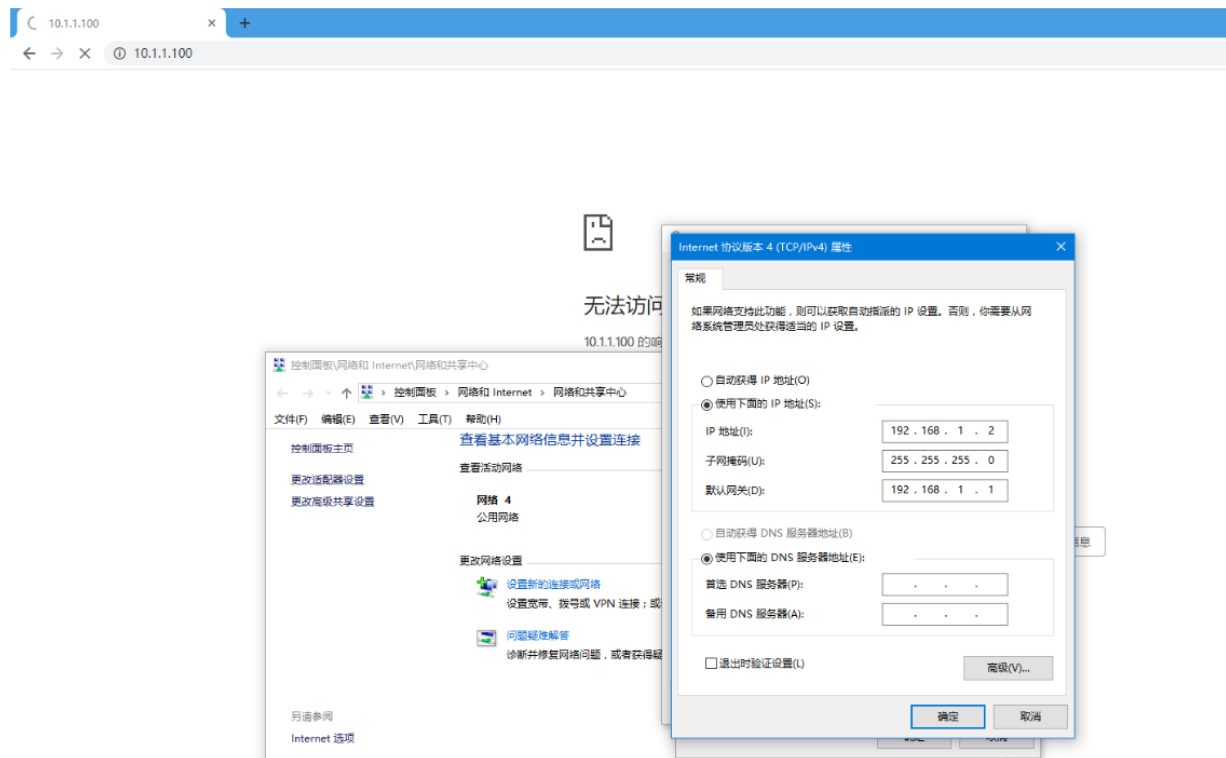




(3)普通员工主机 A、B 分别使用步骤 1 建立的用户名登录 FTP 服务器,并通过 http://10.1.1.100 访问 WWW 服务器,在设定时间段内是否能登录和访问(登录 FTP 时分别通过 DOS 命令与浏览器方式,结合捕获报文分析)?

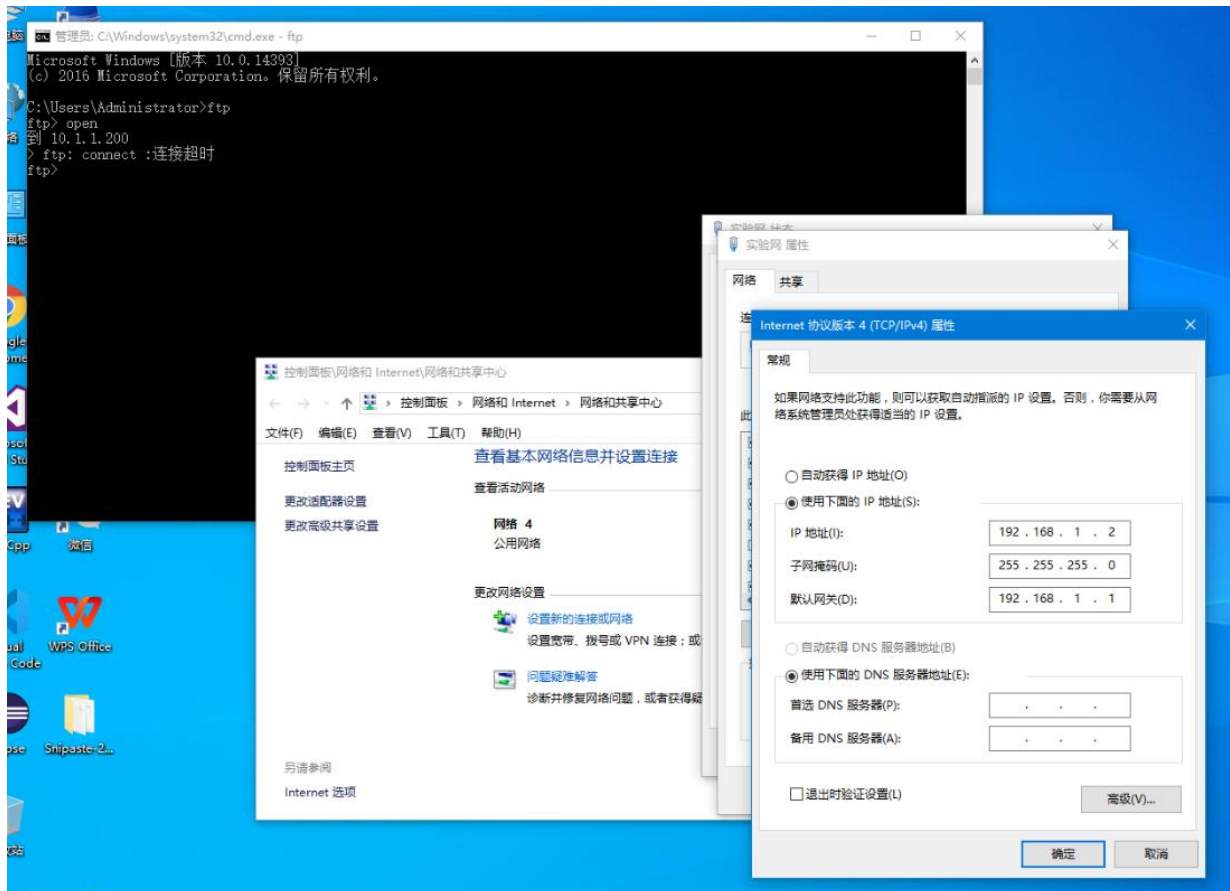
A、B 具有对称性,所以此题以 A 为研究对象即可。

访问 WWW 服务器:





用 DOS 命令登录 FTP 服务器：



用浏览器方式登录 FTP 服务器：



无法访问此网站

无法访问 <http://10.1.1.200/>。

尝试运行 Windows 网络诊断。

ERR_ADDRESS_UNREACHABLE

重新加载

如果网络支持此功能，则可以获取自动指派的 IP 设置。否则，你需要从网络系统管理员处获得适当的 IP 设置。

☐ 自动获得 IP 地址(O)

☒ 使用下面的 IP 地址(S):

IP 地址(I): 192 . 168 . 1 . 2

子网掩码(U): 255 . 255 . 255 . 0

默认网关(D): 192 . 168 . 1 . 1

☐ 自动获得 DNS 服务器地址(B)

☒ 使用下面的 DNS 服务器地址(E):

首选 DNS 服务器(P): . . .

备用 DNS 服务器(A): . . .

具体的捕获报文分析可见(5)登录失败时的报文分析。

(4)改变路由器系统时间段,在其他时间段执行(2)~(3)的测试。

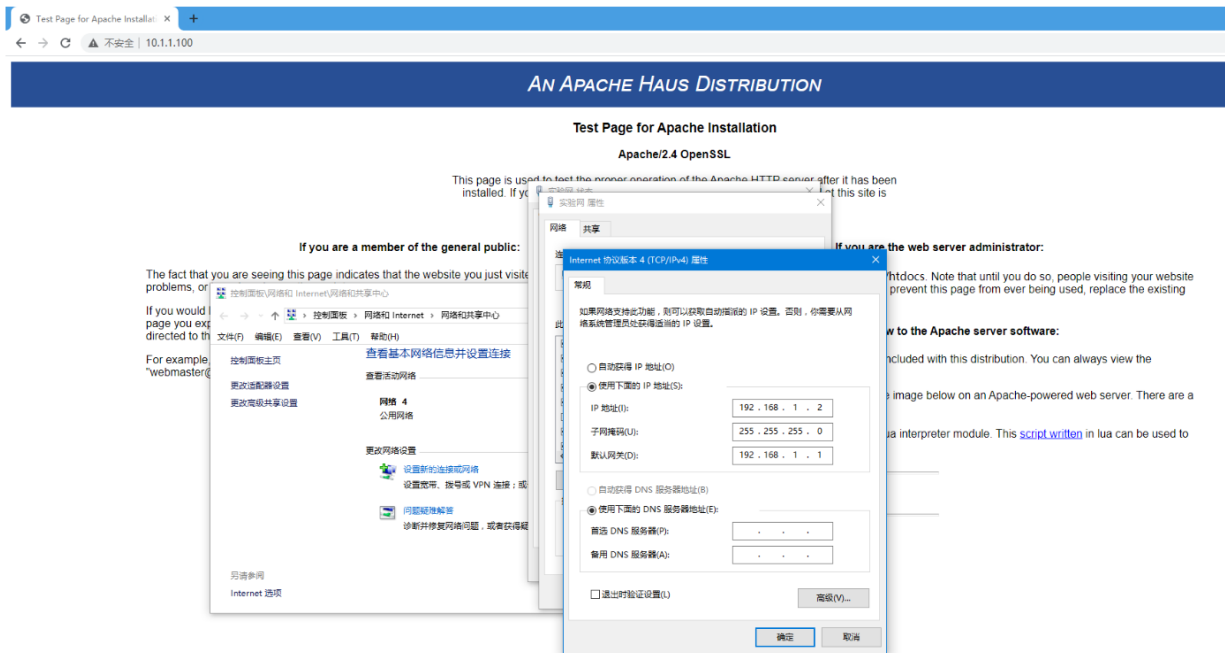
将时间修改为下班时间：

```
Ruijie>enable
Ruijie#clock set 17:00:00 6 5 2021
Ruijie#Jun 5 17:00:00: %SYS-6-CLOCKUPDATE: system clock has been updated to 17:00:00 UTC Sat Jun 5 2021.
```

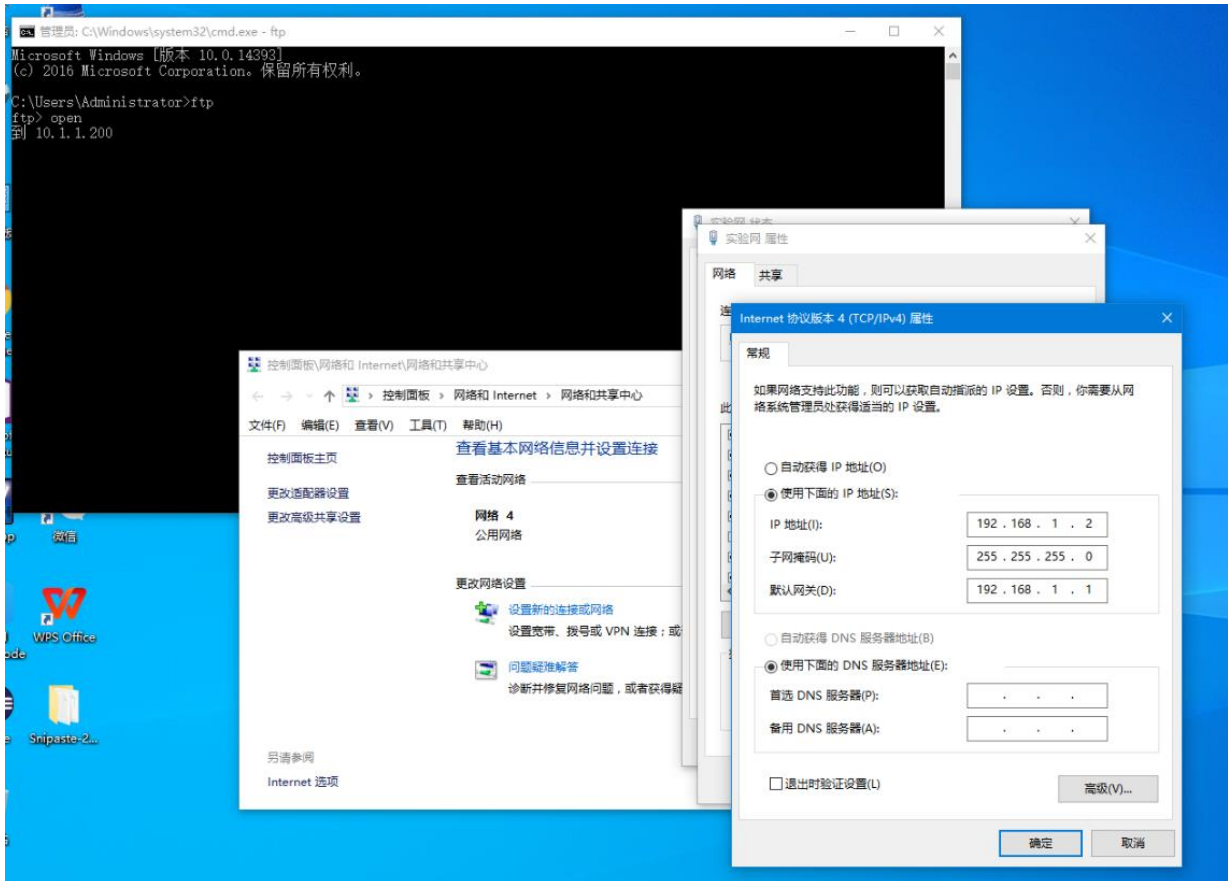


结果如下：

访问 WWW 服务器：

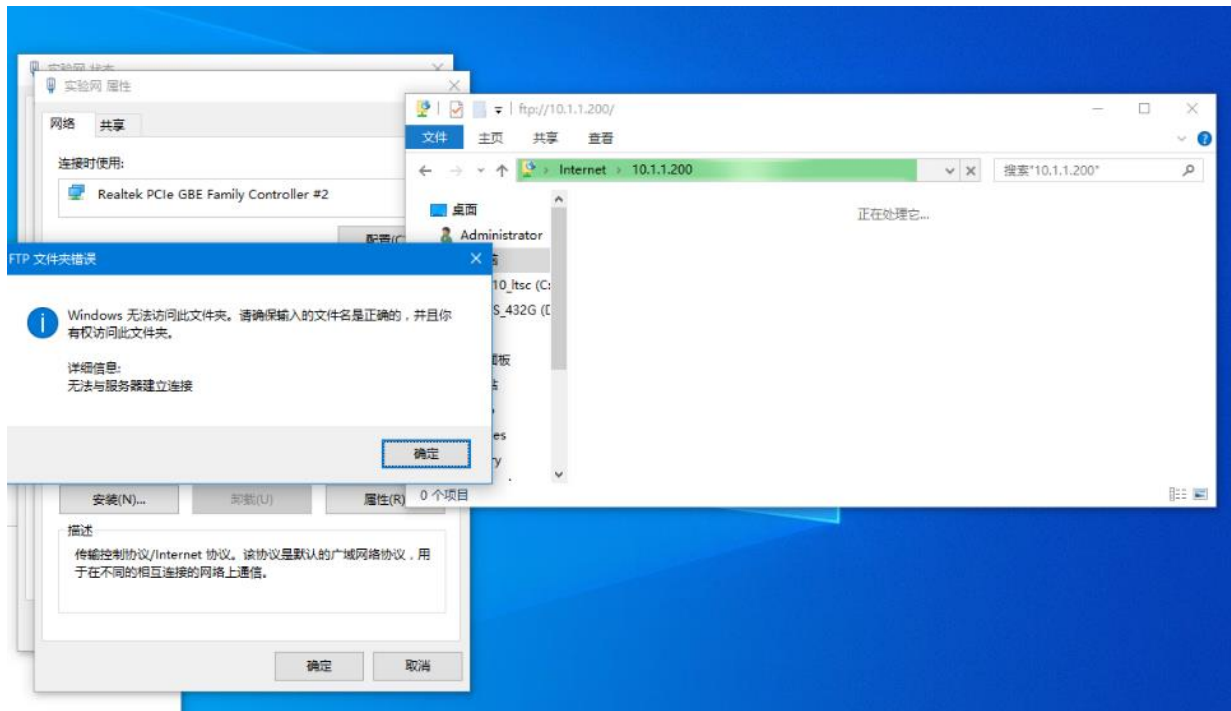


用 DOS 命令登录 FTP 服务器：





用浏览器方式登录 FTP 服务器：



(5)捕获主机访问服务器时的数据包，并进行分析。

经理访问 WWW 服务器：

1	0.000000	10.1.1.200	10.1.1.255	UDP	1486 49859 → 1689 Len=1440
2	0.271541	192.168.1.254	10.1.1.100	TCP	70 4053 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
3	0.271592	10.1.1.100	192.168.1.254	TCP	66 80 → 4053 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
4	0.271644	192.168.1.254	10.1.1.100	TCP	70 4054 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
5	0.271664	10.1.1.100	192.168.1.254	TCP	66 80 → 4054 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
6	0.271909	192.168.1.254	10.1.1.100	TCP	64 4053 → 80 [ACK] Seq=1 Ack=1 Win=525568 Len=0
7	0.271909	192.168.1.254	10.1.1.100	TCP	64 4054 → 80 [ACK] Seq=1 Ack=1 Win=525568 Len=0
8	0.272088	192.168.1.254	10.1.1.100	HTTP	599 GET / HTTP/1.1
9	0.272426	10.1.1.100	192.168.1.254	HTTP	344 HTTP/1.1 304 Not Modified
10	0.323572	192.168.1.254	10.1.1.100	TCP	64 4054 → 80 [ACK] Seq=542 Ack=291 Win=525056 Len=0
11	2.187901	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
12	5.079192	192.168.1.254	10.1.1.100	HTTP	599 GET / HTTP/1.1
13	5.079530	10.1.1.100	192.168.1.254	HTTP	317 HTTP/1.1 304 Not Modified
14	5.130383	192.168.1.254	10.1.1.100	TCP	64 4054 → 80 [ACK] Seq=1083 Ack=554 Win=524800 Len=0
15	5.818349	RuijieNe_15:57:e8	LLDP Multicast	LLDP	395 MA/58:69:6c:15:57:e8 MA/58:69:6c:15:57:e8 121 SysN=19-S5750-2 SysD=Ruijie Layer 3 FULL
16	8.528793	10.1.1.200	10.1.1.255	UDP	1486 49859 → 1689 Len=1440
17	10.079889	10.1.1.100	192.168.1.254	TCP	54 80 → 4054 [FIN, ACK] Seq=554 Ack=1083 Win=525056 Len=0
18	10.080317	192.168.1.254	10.1.1.100	TCP	64 4054 → 80 [ACK] Seq=1083 Ack=555 Win=524800 Len=0
19	10.712435	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
20	17.061130	10.1.1.200	10.1.1.255	UDP	1486 49859 → 1689 Len=1440
21	19.247573	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
22	25.087230	192.168.1.254	10.1.1.100	TCP	64 4054 → 80 [FIN, ACK] Seq=1083 Ack=555 Win=524800 Len=0
23	25.087230	192.168.1.254	10.1.1.100	TCP	70 4060 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
24	25.087318	10.1.1.100	192.168.1.254	TCP	54 80 → 4054 [ACK] Seq=555 Ack=1084 Win=525056 Len=0
25	25.087386	10.1.1.100	192.168.1.254	TCP	66 80 → 4060 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
26	25.087428	192.168.1.254	10.1.1.100	HTTP	599 GET / HTTP/1.1
27	25.087722	192.168.1.254	10.1.1.100	TCP	64 4060 → 80 [ACK] Seq=1 Ack=1 Win=525568 Len=0
28	25.088205	10.1.1.100	192.168.1.254	HTTP	344 HTTP/1.1 304 Not Modified
29	25.139403	192.168.1.254	10.1.1.100	TCP	64 4053 → 80 [ACK] Seq=542 Ack=291 Win=525056 Len=0
30	35.501844	10.1.1.200	10.1.1.255	UDP	1486 49859 → 1689 Len=1440

可以看到建立 TCP 连接和 HTTP 传输的过程

员工在工作时间访问 WWW 服务器：

服务器抓包：

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	RuijieNe_15:57:e8	LLDP Multicast	LLDP	395	MA/58:69:6c:15:57:e8 MA/58:69:6c:15:57:e8 121 SysN=19-S5750-2
2	0.167406	10.1.1.200	10.1.1.255	UDP	1486	49859 → 1689 Len=1440
3	2.397446	10.1.1.100	10.1.1.255	UDP	1482	52026 → 1689 Len=1440
4	8.698374	10.1.1.200	10.1.1.255	UDP	1486	49859 → 1689 Len=1440
5	10.934468	10.1.1.100	10.1.1.255	UDP	1482	52026 → 1689 Len=1440
6	12.137840	fe80::8895:e153:1c6...	ff02::1:2	DHCPv6	161	Solicit XID: 0x0d9e83 CID: 000100012723eb7880c16ee3ca42



客户端抓包：

117 9.693501	192.168.1.2	10.1.1.100	TCP	66 2367 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
118 9.694113	192.168.1.2	10.1.1.100	TCP	66 2368 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
120 9.944494	192.168.1.2	10.1.1.100	TCP	66 2369 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
163 12.693392	192.168.1.2	10.1.1.100	TCP	66 [TCP Retransmission] 2367 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
164 12.695392	192.168.1.2	10.1.1.100	TCP	66 [TCP Retransmission] 2368 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
167 12.944705	192.168.1.2	10.1.1.100	TCP	66 [TCP Retransmission] 2369 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
206 18.694177	192.168.1.2	10.1.1.100	TCP	62 [TCP Retransmission] 2367 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
207 18.695205	192.168.1.2	10.1.1.100	TCP	62 [TCP Retransmission] 2368 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM=1
211 18.945630	192.168.1.2	10.1.1.100	TCP	62 [TCP Retransmission] 2369 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM=1

可以看到由于 ACL 配置，数据包在路由器被阻拦，客户端多次进行了 TCP 重传，与服务器建立连接失败，服务器并未收到任何 HTTP 和 TCP 数据包

员工在非工作时间访问 WWW 服务器：

服务器抓包：

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.1.1.200	10.1.1.255	UDP	1486	49859 → 1689 Len=1440
2	0.398448	Ruijie# 15:57:08	LDP Multicast	LDP	395	MA/58:09:6c:15:57:08 MA/58:09:6c:15:57:08 121 SysN=19-55750-2 SysD=Ruijie Layer 3 FULL Gigab
3	2.230657	10.1.1.100	10.1.1.255	UDP	1482	52026 → 1689 Len=1440
4	3.759727	fe80::a1f9:41a9:bf4...	ff02::1:2	DHCPv6	157	Solicit XID: 0xbcd6da CID: 000100012723eb7880c16ee3ca42
5	8.082291	192.168.1.2	10.1.1.100	TCP	70	2416 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
6	8.082405	10.1.1.100	192.168.1.2	TCP	66	80 → 2416 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
7	8.082696	192.168.1.2	10.1.1.100	TCP	64	2416 → 80 [ACK] Seq=1 Ack=1 Win=525568 Len=0
8	8.082786	192.168.1.2	10.1.1.100	HTTP	505	GET / HTTP/1.1
9	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=1 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
10	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=1461 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
11	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=2921 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
12	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=4381 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
13	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=5841 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
14	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=7301 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
15	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=8761 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
16	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=10221 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
17	8.083798	10.1.1.100	192.168.1.2	TCP	1514	80 → 2416 [ACK] Seq=11681 Ack=448 Win=525568 Len=1460 [TCP segment of a reassembled PDU]
18	8.083798	10.1.1.100	192.168.1.2	HTTP	1076	HTTP/1.1 200 OK (text/html)
19	8.084419	192.168.1.2	10.1.1.100	TCP	64	2416 → 80 [ACK] Seq=448 Ack=14163 Win=525568 Len=0
20	8.531697	10.1.1.200	10.1.1.255	UDP	1486	49859 → 1689 Len=1440
21	8.566869	192.168.1.2	10.1.1.100	TCP	70	2418 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
22	8.566962	10.1.1.100	192.168.1.2	TCP	66	80 → 2418 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
23	8.567125	192.168.1.2	10.1.1.100	TCP	64	2418 → 80 [ACK] Seq=1 Ack=1 Win=525568 Len=0
24	8.613134	192.168.1.2	10.1.1.100	HTTP	425	GET /favicon.ico HTTP/1.1
25	8.613824	10.1.1.100	192.168.1.2	HTTP	480	HTTP/1.1 404 Not Found (text/html)
26	8.664170	192.168.1.2	10.1.1.100	TCP	64	2416 → 80 [ACK] Seq=815 Ack=14589 Win=525056 Len=0
27	10.767061	10.1.1.100	10.1.1.255	UDP	1482	52026 → 1689 Len=1440
28	13.613374	10.1.1.100	192.168.1.2	TCP	54	80 → 2416 [FIN, ACK] Seq=14589 Ack=815 Win=525056 Len=0
29	13.613816	192.168.1.2	10.1.1.100	TCP	64	2416 → 80 [ACK] Seq=815 Ack=14590 Win=525056 Len=0
30	14.338851	192.168.1.2	10.1.1.100	TCP	64	2416 → 80 [FIN, ACK] Seq=815 Ack=14590 Win=525056 Len=0
31	14.338917	10.1.1.100	192.168.1.2	TCP	54	80 → 2416 [ACK] Seq=14590 Ack=816 Win=525056 Len=0
32	17.062903	10.1.1.200	10.1.1.255	UDP	1486	49859 → 1689 Len=1440
33	19.208652	10.1.1.100	10.1.1.255	UDP	1482	52026 → 1689 Len=1440
34	19.760599	fe80::a1f9:41a9:bf4...	ff02::1:2	DHCPv6	157	Solicit XID: 0xbcd6da CID: 000100012723eb7880c16ee3ca42
35	23.840193	192.168.1.2	10.1.1.100	TCP	70	2444 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
36	23.840193	192.168.1.2	10.1.1.100	HTTP	599	GET / HTTP/1.1
37	23.840287	10.1.1.100	192.168.1.2	TCP	66	80 → 2444 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
38	23.840529	192.168.1.2	10.1.1.100	TCP	64	2444 → 80 [ACK] Seq=1 Ack=1 Win=525568 Len=0
39	23.841022	10.1.1.100	192.168.1.2	HTTP	344	HTTP/1.1 304 Not Modified
40	23.891939	192.168.1.2	10.1.1.100	TCP	64	2418 → 80 [ACK] Seq=542 Ack=291 Win=525056 Len=0
41	25.512118	192.168.1.2	10.1.1.100	HTTP	599	GET / HTTP/1.1

客户端抓包：

No.	Time	Source	Destination	Protocol	Length	Info
33	4.418623	00:08:99:00:0a:49	Ruijie# 27:b8:19	ARP	42	who has 192.168.1.1? Tell 192.168.1.2
34	4.428473	Ruijie# 27:b8:19	00:08:99:00:0a:49	ARP	60	192.168.1.1 is at 58:69:6c:27:b8:19
36	4.448382	192.168.1.2	10.1.1.100	HTTP	595	GET / HTTP/1.1
39	4.449511	10.1.1.100	192.168.1.2	HTTP	344	HTTP/1.1 304 Not Modified
47	6.120095	192.168.1.2	10.1.1.100	HTTP	595	GET / HTTP/1.1
48	6.121172	10.1.1.100	192.168.1.2	HTTP	317	HTTP/1.1 304 Not Modified
52	6.664010	192.168.1.2	10.1.1.100	HTTP	595	GET / HTTP/1.1
53	6.665126	10.1.1.100	192.168.1.2	HTTP	317	HTTP/1.1 304 Not Modified
3	0.200301	192.168.1.2	183.192.173.203	TCP	66	2438 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
6	0.470191	192.168.1.2	120.241.16.15	TCP	62	2424 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
8	0.501671	192.168.1.2	183.192.173.203	TCP	66	2439 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
10	0.903031	192.168.1.2	183.192.173.203	TCP	66	2440 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
12	1.103069	192.168.1.2	183.192.199.230	TCP	66	2441 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
16	1.906138	192.168.1.2	183.192.173.203	TCP	66	2442 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
17	1.996324	192.168.1.2	183.192.173.203	TCP	66	2437 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
19	2.609930	192.168.1.2	183.192.173.203	TCP	66	2443 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
20	3.200567	192.168.1.2	183.192.173.203	TCP	66	[TCP Retransmission] 2438 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
21	3.501644	192.168.1.2	183.192.173.203	TCP	66	[TCP Retransmission] 2439 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
22	3.982863	192.168.1.2	183.192.173.203	TCP	66	[TCP Retransmission] 2440 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.984322	192.168.1.2	183.192.173.203	TCP	62	2428 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
24	3.984343	192.168.1.2	183.192.173.203	TCP	62	2427 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
25	3.984410	192.168.1.2	183.192.173.203	TCP	62	2425 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
26	3.984435	192.168.1.2	183.192.173.203	TCP	62	2426 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
27	3.985363	192.168.1.2	183.192.173.203	TCP	62	2431 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
28	3.986441	192.168.1.2	183.192.173.203	TCP	62	2429 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
29	3.986484	192.168.1.2	183.192.173.203	TCP	62	2430 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
30	4.103616	192.168.1.2	183.192.199.230	TCP	66	[TCP Retransmission] 2441 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
31	4.185379	192.168.1.2	183.192.173.203	TCP	62	2432 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
32	4.387320	192.168.1.2	183.192.199.230	TCP	62	2433 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
35	4.448282	192.168.1.2	10.1.1.100	TCP	66	2444 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
37	4.448770	10.1.1.100	192.168.1.2	TCP	66	80 → 2444 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
38	4.448794	192.168.1.2	10.1.1.100	TCP	54	2444 → 80 [ACK] Seq=1 Ack=1 Win=525568 Len=0
40	4.509033	192.168.1.2	10.1.1.100	TCP	54	2418 → 80 [ACK] Seq=542 Ack=291 Win=2051 Len=0
41	4.980640	192.168.1.2	183.192.173.203	TCP	66	[TCP Retransmission] 2442 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
42	5.189722	192.168.1.2	183.192.173.203	TCP	62	2435 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
43	5.190775	192.168.1.2	183.192.173.203	TCP	62	2434 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
45	5.609980	192.168.1.2	183.192.173.203	TCP	66	[TCP Retransmission] 2443 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
46	5.792890	192.168.1.2	183.192.173.203	TCP	62	2436 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
50	6.171303	192.168.1.2	10.1.1.100	TCP	54	2418 → 80 [ACK] Seq=1083 Ack=554 Win=2050 Len=0
54	6.715373	192.168.1.2	10.1.1.100	TCP	54	2418 → 80 [ACK] Seq=1624 Ack=817 Win=2049 Len=0
55	7.229206	192.168.1.2	120.241.16.15	TCP	66	2445 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
56	7.995902	192.168.1.2	183.192.173.203	TCP	62	[TCP Retransmission] 2437 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1

可以看到客户端访问 WWW 服务器成功



经理通过 DOS 访问 FTP 服务器：

1 0.000000	00:88:99:00:0b:a7	RuijieNe 27:b8:1a	ARP	42 Who has 10.1.1.1? Tell 10.1.1.200
2 0.000388	RuijieNe 27:b8:1a	00:88:99:00:0b:a7	ARP	60 10.1.1.1 is at 58:69:6c:27:b8:1a
3 0.813366	10.1.1.200	10.1.1.255	UDP	1482 49859 → 1689 Len=1440
4 0.914548	10.1.1.200	183.192.199.230	TCP	66 8455 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
5 0.914882	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
6 1.747549	fe80::8895:e153:1c6...	ff02::1:2	DHCPv6	157 Solicit XID: 0x007412 CID: 000100012723eb7880c16ee3ca42
7 1.325691	192.168.1.254	10.1.1.200	TCP	66 4080 → 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=1 SACK_PERM=1
8 2.325800	10.1.1.200	192.168.1.254	TCP	66 21 → 4080 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
9 2.326115	192.168.1.254	10.1.1.200	TCP	60 4080 → 21 [ACK] Seq=1 Ack=1 Win=8192 Len=0
10 2.326566	10.1.1.200	192.168.1.254	FTP	197 Response: 220-FileZilla Server 0.9.60 beta
11 2.332377	192.168.1.254	10.1.1.200	FTP	68 Request: OPTS UTF8 ON
12 2.332505	10.1.1.200	192.168.1.254	FTP	118 Response: 202 UTF8 mode is always enabled. No need to send this command.
13 2.383686	192.168.1.254	10.1.1.200	TCP	60 4080 → 21 [ACK] Seq=15 Ack=208 Win=7985 Len=0
14 2.997515	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
15 5.909537	192.168.1.254	10.1.1.200	FTP	70 Request: USER localhost
16 5.909795	10.1.1.200	192.168.1.254	FTP	91 Response: 331 Password required for localhost
17 5.960015	192.168.1.254	10.1.1.200	TCP	60 4080 → 21 [ACK] Seq=31 Ack=245 Win=7948 Len=0
18 6.913948	10.1.1.200	183.192.199.230	TCP	62 [TCP Retransmission] 8455 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
19 6.914272	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
20 8.048638	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
21 8.048908	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
22 8.558298	192.168.1.254	10.1.1.200	FTP	67 Request: PASS 123456
23 8.558774	10.1.1.200	192.168.1.254	FTP	69 Response: 230 Logged on
24 8.560475	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
25 8.560860	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
26 8.609479	192.168.1.254	10.1.1.200	TCP	60 4080 → 21 [ACK] Seq=44 Ack=260 Win=7933 Len=0
27 9.063420	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
28 9.063678	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
29 9.344747	10.1.1.200	10.1.1.255	UDP	1482 49859 → 1689 Len=1440
30 9.564756	10.1.1.200	120.241.16.15	TCP	66 8456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
31 9.565140	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
32 11.531467	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
33 11.600092	RuijieNe 45:57:e0	LLDP Multicast	LLDP	391 MA/58:69:6c:15:57:e0 MA/58:69:6c:15:57:e0 121 Syn=10.55750-2 SynD=Ruijie Layer 3 FULL
34 12.564992	10.1.1.200	120.241.16.15	TCP	66 [TCP Retransmission] 8456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
35 12.565204	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
36 15.341568	192.168.1.254	10.1.1.200	FTP	81 Request: PORT 192,168,1,254,15,243
37 15.341834	10.1.1.200	192.168.1.254	FTP	83 Response: 200 Port command successful
38 15.343950	192.168.1.254	10.1.1.200	FTP	60 Request: NLST
39 15.344630	10.1.1.200	192.168.1.254	TCP	66 20 → 4083 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
40 15.344682	10.1.1.200	192.168.1.254	FTP	109 Response: 150 Opening data channel for directory listing of "/"
41 15.345091	192.168.1.254	10.1.1.200	TCP	66 4083 → 20 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
42 15.345135	10.1.1.200	192.168.1.254	TCP	54 20 → 4083 [ACK] Seq=1 Ack=1 Win=525568 Len=0
43 15.345296	10.1.1.200	192.168.1.254	FTP-DAL	61 FTP Data: 7 bytes (PORT) (NLST)

可以看到经理通过 DOS 访问 FTP 服务器正常

经理通过文件夹访问 FTP 服务器：

1 0.000000	10.1.1.200	183.192.199.230	TCP	66 8446 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
2 0.000358	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
3 1.040146	192.168.1.254	10.1.1.200	TCP	66 4068 → 21 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=8 SACK_PERM=1
4 1.040248	10.1.1.200	192.168.1.254	TCP	66 21 → 4068 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
5 1.040590	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=1 Ack=1 Win=262144 Len=0
6 1.041066	10.1.1.200	192.168.1.254	FTP	197 Response: 220-FileZilla Server 0.9.60 beta
7 1.041324	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=1 Ack=144 Win=262000 Len=0
8 1.041324	192.168.1.254	10.1.1.200	FTP	70 Request: USER localhost
9 1.041557	10.1.1.200	192.168.1.254	FTP	91 Response: 331 Password required for localhost
10 1.041829	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=17 Ack=181 Win=261960 Len=0
11 1.041829	192.168.1.254	10.1.1.200	FTP	67 Request: PASS 123456
12 1.042211	10.1.1.200	192.168.1.254	FTP	69 Response: 230 Logged on
13 1.042457	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=30 Ack=196 Win=261944 Len=0
14 1.042457	192.168.1.254	10.1.1.200	FTP	68 Request: opts utf8 on
15 1.042662	10.1.1.200	192.168.1.254	FTP	118 Response: 202 UTF8 mode is always enabled. No need to send this command.
16 1.042907	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=44 Ack=260 Win=261880 Len=0
17 1.042907	192.168.1.254	10.1.1.200	FTP	60 Request: syst
18 1.043099	10.1.1.200	192.168.1.254	FTP	86 Response: 215 UNIX emulated by FileZilla
19 1.043358	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=50 Ack=292 Win=261848 Len=0
20 1.043358	192.168.1.254	10.1.1.200	FTP	65 Request: site help
21 1.043562	10.1.1.200	192.168.1.254	FTP	102 Response: 504 Command not implemented for that parameter
22 1.043804	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=61 Ack=340 Win=261800 Len=0
23 1.043804	192.168.1.254	10.1.1.200	FTP	60 Request: PWD
24 1.043989	10.1.1.200	192.168.1.254	FTP	85 Response: 257 "/" is current directory.
25 1.044251	192.168.1.254	10.1.1.200	TCP	60 4068 → 21 [ACK] Seq=66 Ack=371 Win=261768 Len=0
26 1.163317	192.168.1.254	10.1.1.200	TCP	66 4069 → 21 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=8 SACK_PERM=1
27 1.163419	10.1.1.200	192.168.1.254	TCP	66 21 → 4069 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
28 1.163739	192.168.1.254	10.1.1.200	TCP	60 4069 → 21 [ACK] Seq=1 Ack=1 Win=262144 Len=0
29 1.164211	10.1.1.200	192.168.1.254	FTP	197 Response: 220-FileZilla Server 0.9.60 beta
30 1.164469	192.168.1.254	10.1.1.200	TCP	60 4069 → 21 [ACK] Seq=1 Ack=144 Win=262000 Len=0
31 1.164469	192.168.1.254	10.1.1.200	FTP	70 Request: USER localhost
32 1.164717	10.1.1.200	192.168.1.254	FTP	91 Response: 331 Password required for localhost
33 1.164972	192.168.1.254	10.1.1.200	TCP	60 4069 → 21 [ACK] Seq=17 Ack=181 Win=261960 Len=0
34 1.164972	192.168.1.254	10.1.1.200	FTP	67 Request: PASS 123456
35 1.165352	10.1.1.200	192.168.1.254	FTP	69 Response: 230 Logged on
36 1.165613	192.168.1.254	10.1.1.200	TCP	60 4069 → 21 [ACK] Seq=30 Ack=196 Win=261944 Len=0
37 1.165613	192.168.1.254	10.1.1.200	FTP	68 Request: opts utf8 on
38 1.165807	10.1.1.200	192.168.1.254	FTP	118 Response: 202 UTF8 mode is always enabled. No need to send this command.
39 1.166095	192.168.1.254	10.1.1.200	TCP	60 4069 → 21 [ACK] Seq=44 Ack=260 Win=261880 Len=0
40 1.166095	192.168.1.254	10.1.1.200	FTP	60 Request: syst
41 1.166281	10.1.1.200	192.168.1.254	FTP	86 Response: 215 UNIX emulated by FileZilla
42 1.166505	192.168.1.254	10.1.1.200	TCP	60 4069 → 21 [ACK] Seq=50 Ack=292 Win=261848 Len=0
43 1.166505	192.168.1.254	10.1.1.200	FTP	65 Request: site help
44 1.166674	10.1.1.200	192.168.1.254	FTP	102 Response: 504 Command not implemented for that parameter

可以看到经理通过文件夹直接访问 FTP 服务器也正常。



员工在工作时间利用 DOS 访问 FTP 服务器:

服务器抓包:

1 0.000000	10.1.1.200	183.192.199.230	TCP	62 8483 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
2 0.000334	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
3 1.141651	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
4 1.141911	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
5 1.640115	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
6 1.640375	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
7 2.143340	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
8 2.143709	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
9 2.328538	10.1.1.200	10.1.1.255	UDP	1482 49859 → 1689 Len=1440
10 2.645452	10.1.1.200	120.241.16.15	TCP	66 8484 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
11 2.645822	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
12 4.545042	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
13 4.952760	00:88:99:00:0b:a7	RuijieNe_27:b8:1a	ARP	42 Who has 10.1.1.1? Tell 10.1.1.200
14 4.955657	RuijieNe_27:b8:1a	00:88:99:00:0b:a7	ARP	60 10.1.1.1 is at 58:69:6c:27:b8:1a
15 5.645875	10.1.1.200	120.241.16.15	TCP	66 [TCP Retransmission] 8484 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
16 5.646206	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
17 6.358263	192.168.1.2	10.1.1.200	TCP	66 2339 → 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=1 SACK_PERM=1
18 6.358382	10.1.1.200	192.168.1.2	TCP	66 21 → 2339 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
19 6.358907	192.168.1.2	10.1.1.200	TCP	60 2339 → 21 [ACK] Seq=1 Ack=1 Win=8192 Len=0
20 6.359414	10.1.1.200	192.168.1.2	FTP	197 Response: 220-FileZilla Server 0.9.60 beta
21 6.364168	192.168.1.2	10.1.1.200	FTP	68 Request: OPTS UTF8 ON
22 6.364360	10.1.1.200	192.168.1.2	FTP	118 Response: 202 UTF8 mode is always enabled. No need to send this command.
23 6.415055	192.168.1.2	10.1.1.200	TCP	60 2339 → 21 [ACK] Seq=15 Ack=208 Win=7985 Len=0
24 10.070146	192.168.1.2	10.1.1.200	FTP	70 Request: USER localhost
25 10.070347	10.1.1.200	192.168.1.2	FTP	91 Response: 331 Password required for localhost
26 10.121977	192.168.1.2	10.1.1.200	TCP	60 2339 → 21 [ACK] Seq=31 Ack=245 Win=7948 Len=0
27 10.859619	10.1.1.200	10.1.1.255	UDP	1482 49859 → 1689 Len=1440
28 11.645932	10.1.1.200	120.241.16.15	TCP	62 [TCP Retransmission] 8484 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
29 11.646272	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
30 12.206661	192.168.1.2	10.1.1.200	FTP	67 Request: PASS 123456
31 12.207048	10.1.1.200	192.168.1.2	FTP	69 Response: 230 Logged on
32 12.259361	192.168.1.2	10.1.1.200	TCP	60 2339 → 21 [ACK] Seq=44 Ack=260 Win=7933 Len=0
33 12.688505	10.1.1.200	183.192.199.230	UDP	164 51365 → 8000 Len=122
34 12.688873	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
35 13.077631	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
36 13.189426	10.1.1.200	183.192.199.230	UDP	164 51365 → 8000 Len=122
37 13.189766	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
38 13.690548	10.1.1.200	183.192.199.230	UDP	164 51365 → 8000 Len=122
39 13.690917	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
40 13.790197	192.168.1.2	10.1.1.200	FTP	77 Request: PORT 192,168,1,2,9,38
41 13.790493	10.1.1.200	192.168.1.2	FTP	83 Response: 200 Port command successful
42 13.792564	192.168.1.2	10.1.1.200	FTP	60 Request: NLST
43 13.793281	10.1.1.200	192.168.1.2	TCP	66 20 → 2342 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
44 13.793333	10.1.1.200	192.168.1.2	FTP	109 Response: 150 Opening data channel for directory listing of "/"

客户端抓包:

1 0.000000	192.168.1.2	192.168.1.255	UDP	1482 64859 → 1689 Len=1440
2 1.603879	RuijieNe_15:57:1e	LLDP Multicast	LLDP	390 MA/58:69:6c:15:57:1e MA/58:69:6c:15:57:1e 121 SysN=19-55750-1 SysD=Ruijie Layer 3 FULL Gig
3 1.853899	00:88:99:00:0a:49	RuijieNe_27:b8:19	ARP	42 Who has 192.168.1.1? Tell 192.168.1.2
4 1.858735	RuijieNe_27:b8:19	00:88:99:00:0a:49	ARP	60 192.168.1.1 is at 58:69:6c:27:b8:19
5 1.922291	192.168.1.254	192.168.1.255	UDP	1482 64211 → 1689 Len=1440
6 2.932940	192.168.1.2	183.192.173.203	TCP	62 2336 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
7 3.261773	192.168.1.2	183.192.199.230	TCP	62 2337 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
8 4.404624	192.168.1.2	120.241.16.15	UDP	164 52318 → 8000 Len=122
9 4.901137	192.168.1.2	120.241.16.15	UDP	164 52318 → 8000 Len=122
10 5.403112	192.168.1.2	120.241.16.15	UDP	164 52318 → 8000 Len=122
11 5.906296	192.168.1.2	120.241.16.15	TCP	66 2338 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
12 8.527185	192.168.1.2	192.168.1.255	UDP	1482 64859 → 1689 Len=1440
13 8.905789	192.168.1.2	120.241.16.15	TCP	66 [TCP Retransmission] 2338 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
14 10.455483	192.168.1.254	192.168.1.255	UDP	1482 64211 → 1689 Len=1440
15 10.729227	192.168.1.2	10.1.1.200	TCP	66 2339 → 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=1 SACK_PERM=1
16 10.729808	10.1.1.200	192.168.1.2	TCP	66 21 → 2339 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
17 10.729851	192.168.1.2	10.1.1.200	TCP	54 2339 → 21 [ACK] Seq=1 Ack=1 Win=8192 Len=0
18 10.730726	10.1.1.200	192.168.1.2	FTP	197 Response: 220-FileZilla Server 0.9.60 beta
19 10.735286	192.168.1.2	10.1.1.200	FTP	68 Request: OPTS UTF8 ON
20 10.735643	10.1.1.200	192.168.1.2	FTP	118 Response: 202 UTF8 mode is always enabled. No need to send this command.
21 10.786077	192.168.1.2	10.1.1.200	TCP	54 2339 → 21 [ACK] Seq=15 Ack=208 Win=7985 Len=0
22 12.666176	fe80::449c:fb59:d211::ff02::1:2	DHCPv6	DHCPv6	157 Solicit XID: 0x07f6e0 CID: 000100012723eb7880c16ee3ca42
23 13.666271	fe80::449c:fb59:d211::ff02::1:2	DHCPv6	DHCPv6	157 Solicit XID: 0x07f6e0 CID: 000100012723eb7880c16ee3ca42
24 13.692729	192.168.1.2	120.241.16.15	TCP	66 2340 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
25 14.441204	192.168.1.2	10.1.1.200	FTP	70 Request: USER localhost
26 14.441657	10.1.1.200	192.168.1.2	FTP	91 Response: 331 Password required for localhost
27 14.492971	192.168.1.2	10.1.1.200	TCP	54 2339 → 21 [ACK] Seq=31 Ack=245 Win=7948 Len=0
28 14.906279	192.168.1.2	120.241.16.15	TCP	62 [TCP Retransmission] 2340 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
29 15.667038	fe80::449c:fb59:d211::ff02::1:2	DHCPv6	DHCPv6	157 Solicit XID: 0x07f6e0 CID: 000100012723eb7880c16ee3ca42
30 16.039443	192.168.1.2	183.192.199.230	UDP	164 52318 → 8000 Len=122
31 16.541549	192.168.1.2	183.192.199.230	UDP	164 52318 → 8000 Len=122
32 16.577726	192.168.1.2	10.1.1.200	FTP	67 Request: PASS 123456
33 16.578441	10.1.1.200	192.168.1.2	FTP	69 Response: 230 Logged on
34 16.630351	192.168.1.2	10.1.1.200	TCP	54 2339 → 21 [ACK] Seq=44 Ack=260 Win=7933 Len=0
35 16.692831	192.168.1.2	120.241.16.15	TCP	66 [TCP Retransmission] 2340 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
36 17.042455	192.168.1.2	183.192.199.230	UDP	164 52318 → 8000 Len=122
37 17.066327	192.168.1.2	192.168.1.255	UDP	1482 64859 → 1689 Len=1440
38 17.545115	192.168.1.2	183.192.199.230	TCP	66 2341 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
39 18.161138	192.168.1.2	10.1.1.200	FTP	77 Request: PORT 192,168,1,2,9,38
40 18.161887	10.1.1.200	192.168.1.2	FTP	83 Response: 200 Port command successful
41 18.163636	192.168.1.2	10.1.1.200	FTP	60 Request: NLST
42 18.164592	10.1.1.200	192.168.1.2	TCP	66 20 → 2342 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
43 18.164592	10.1.1.200	192.168.1.2	FTP	109 Response: 150 Opening data channel for directory listing of "/"

可以看到员工在工作时间利用 DOS 访问 FTP 服务器正常, 但用浏览器方式访问时并不能访问到 FTP 服务器。在这里我们猜测是因为 ACL 配置只允许了 TCP 协议的传输层协议, 但利用浏览器方式访问是基于其他协议访问, 所以导致访问不了。但经理的 IP 地



址设置的是允许一切访问，所以经理可以通过浏览器方式访问到 FTP 服务器。

员工在非工作时间利用 DOS 访问 FTP 服务器：

服务器抓包：

1	0.000000	10.1.1.200	183.192.199.230	TCP	66 8539 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
2	0.000365	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
3	0.071921	10.1.1.200	120.241.16.15	TCP	62 8538 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
4	0.072278	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
5	0.155722	10.1.1.200	183.192.173.203	UDP	508 63741 → 8000 Len=466
6	0.156061	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
7	1.108869	10.1.1.200	183.192.173.203	UDP	284 55544 → 8000 Len=242
8	1.109231	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
9	1.248100	10.1.1.200	183.192.173.203	TCP	66 8540 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
10	1.248330	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
11	1.260253	10.1.1.200	183.192.173.203	UDP	508 63741 → 8000 Len=466
12	1.260513	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
13	2.213370	10.1.1.200	183.192.173.203	UDP	284 55544 → 8000 Len=242
14	2.213707	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
15	2.307205	10.1.1.200	10.1.1.255	UDP	1482 49859 → 1689 Len=1440
16	3.217460	10.1.1.200	183.192.173.203	UDP	284 55544 → 8000 Len=242
17	3.217833	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
18	4.321107	10.1.1.200	183.192.173.203	TCP	66 8541 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
19	4.321477	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
20	4.544321	10.1.1.100	10.1.1.255	UDP	1482 52026 → 1689 Len=1440
21	4.838081	00:88:99:00:0b:a7	RuijieNe_27:b8:1a	ARP	42 Who has 10.1.1.1? Tell 10.1.1.200
22	4.839121	RuijieNe_27:b8:1a	00:88:99:00:0b:a7	ARP	60 10.1.1.1 is at 58:69:6c:27:b8:1a
23	6.000004	10.1.1.200	183.192.199.230	TCP	62 [TCP Retransmission] 8539 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
24	6.000354	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
25	7.138977	10.1.1.200	120.241.16.15	TCP	66 8542 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
26	7.139201	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
27	7.139201	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
28	7.139422	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
29	7.248319	10.1.1.200	183.192.173.203	TCP	62 [TCP Retransmission] 8540 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
30	7.248650	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
31	7.321340	10.1.1.200	183.192.173.203	TCP	66 [TCP Retransmission] 8541 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
32	7.321692	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
33	7.747436	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
34	7.747768	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
35	8.248229	10.1.1.200	120.241.16.15	UDP	164 51365 → 8000 Len=122
36	8.248605	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
37	8.749511	10.1.1.200	120.241.16.15	TCP	66 8543 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
38	8.749879	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
39	10.138513	10.1.1.200	120.241.16.15	TCP	66 [TCP Retransmission] 8542 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
40	10.138880	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)
41	10.838429	10.1.1.200	10.1.1.255	UDP	1482 49859 → 1689 Len=1440
42	11.749837	10.1.1.200	120.241.16.15	TCP	66 [TCP Retransmission] 8543 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
43	11.750171	10.1.1.1	10.1.1.200	ICMP	70 Destination unreachable (Network unreachable)

可以看到捕获到很多路由器发给服务器的数据包，内容为 Destination unreachable(Network unreachable)，且没有捕获到 ftp 数据包。

客户端抓包：

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.2	192.168.1.255	UDP	1482	64859 → 1689 Len=1440
2	0.672012	192.168.1.2	183.192.199.230	TCP	62	2448 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
3	1.298761	192.168.1.2	10.1.1.200	TCP	66	2449 → 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=1 SACK_PERM=1
4	1.777776	192.168.1.2	120.241.16.15	UDP	164	52318 → 8000 Len=122
5	1.921781	192.168.1.254	192.168.1.255	UDP	1482	64211 → 1689 Len=1440
6	2.311484	192.168.1.2	120.241.16.15	UDP	164	52318 → 8000 Len=122
7	2.913699	192.168.1.2	120.241.16.15	UDP	164	52318 → 8000 Len=122
8	3.416874	192.168.1.2	120.241.16.15	TCP	66	2450 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
9	4.299091	192.168.1.2	10.1.1.200	TCP	66	[TCP Retransmission] 2449 → 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=1 SACK_PERM=1
10	5.130597	192.168.1.254	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
11	5.228126	00:88:99:00:0a:49	RuijieNe_27:b8:19	ARP	42	Who has 192.168.1.1? Tell 192.168.1.2
12	5.228536	RuijieNe_27:b8:19	00:88:99:00:0a:49	ARP	60	192.168.1.1 is at 58:69:6c:27:b8:19
13	6.130600	192.168.1.254	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
14	6.418292	192.168.1.2	120.241.16.15	TCP	66	[TCP Retransmission] 2450 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
15	7.131348	192.168.1.254	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
16	8.131393	192.168.1.254	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
17	8.532550	192.168.1.2	192.168.1.255	UDP	1482	64859 → 1689 Len=1440
18	9.912502	169.254.129.97	169.254.255.255	NBNS	92	Name query NB WPAD<00>
19	10.299315	192.168.1.2	10.1.1.200	TCP	62	[TCP Retransmission] 2449 → 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM=1
20	10.461243	192.168.1.254	192.168.1.255	UDP	1482	64211 → 1689 Len=1440
21	12.418468	192.168.1.2	120.241.16.15	TCP	62	[TCP Retransmission] 2450 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
22	13.554554	192.168.1.2	183.192.199.230	UDP	164	52318 → 8000 Len=122
23	14.056709	192.168.1.2	183.192.199.230	UDP	164	52318 → 8000 Len=122
24	14.557832	192.168.1.2	183.192.199.230	UDP	164	52318 → 8000 Len=122
25	15.059830	192.168.1.2	183.192.199.230	TCP	66	2451 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
26	15.227490	00:88:99:00:0a:49	RuijieNe_27:b8:19	ARP	42	Who has 192.168.1.1? Tell 192.168.1.2
27	15.228784	RuijieNe_27:b8:19	00:88:99:00:0a:49	ARP	60	192.168.1.1 is at 58:69:6c:27:b8:19
28	17.061837	192.168.1.2	192.168.1.255	UDP	1482	64859 → 1689 Len=1440
29	18.060266	192.168.1.2	183.192.199.230	TCP	66	[TCP Retransmission] 2451 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1
30	18.985821	192.168.1.254	192.168.1.255	UDP	1482	64211 → 1689 Len=1440
31	24.060474	192.168.1.2	183.192.199.230	TCP	62	[TCP Retransmission] 2451 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
32	24.484233	RuijieNe_15:57:1e	LLDP_Multicast	LLDP	390	MA/58:69:6c:15:57:1e MA/58:69:6c:15:57:1e 121 SysN=19-S5750-1 SysD=Ruijie Layer 3 FULL
33	25.297089	192.168.1.2	120.241.16.15	UDP	164	52318 → 8000 Len=122
34	25.597083	192.168.1.2	192.168.1.255	UDP	1482	64859 → 1689 Len=1440
35	25.902376	192.168.1.2	120.241.16.15	UDP	164	52318 → 8000 Len=122
36	26.403455	192.168.1.2	120.241.16.15	UDP	164	52318 → 8000 Len=122
37	26.905004	192.168.1.2	120.241.16.15	TCP	66	2452 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2 SACK_PERM=1

可以看到捕获到许多 TCP 重传的包，并没有捕获到 FTP 数据包，连接服务器失败



计算机网络实验报告

ACL 应用广泛,例如在 NAT、IPv4-IPv6 地址翻译、VPN 技术,QoS 中都使用了 ACL,因此需要熟练掌握。

学号	学生	自评分
18329015	郝裕玮	100
18325071	张闯	100
19335153	马淙升	100